Aryan Patel - Business Forecasting Project - Forecasting Global Monthly Temperatures

Forecast and Conclusion : Using RMSE and MAE the best performing model on the test dataset was the Holt-Winters Model. The Holt-Winters model forecasted the following global monthly temperatures for the test set which included the monthly temperatures from 2012-10-01 to 2013-09-01. October 2012: Holt-Winters predicted 68.3 while the actual value was 68.2.November 2012: Holt-Winters predicted 64.1 while the actual value was 64.6. December 2012: Holt-Winters predicted 60.2 while the actual value was 59.5. January 2013: Holt-Winters predicted 59.0 while the actual value was 59.1. February 2013: Holt-Winters predicted 60.5 while the actual value was 60.6. March 2013: Holt-Winters predicted 63.9 while the actual value was 63.0. April 2013: Holt-Winters predicted 67.3 while the actual value was 66.9. May 2013: Holt-Winters predicted 70.8 while the actual value was 70.7. June 2013: Holt-Winters predicted 73.1 while the actual value was 72.9. July 2013: Holt-Winters predicted 74.4 while the actual value was 74.2. August 2013: Holt-Winters predicted 74.2 while the actual value was 73.9. September 2013: Holt-Winters predicted 72.0 while the actual value was 77.1. Holt Winters had RMSE of 1.51 and MAE of 0.71. For the Holt Winters model : Alpha was 0.34 , Beta was 0.0013, and Gamma was 0.0001. The only other model that came close in the test performance is the ARIMA / SARIMA model which had MAE of 0.71 but RMSE of 1.60 on the test set. The final arima model from running auto arima on the training set was found to be ARIMA(3,0,0)(0,1,2). Due to the large difference in RMSE but the MAE being almost identical in the test set I chose the Holt Winters model as the best performing model on the test set. The implication is that now we can accurately predict global monthly temperatures with less than 1 degree of error. This allows for scientists , policymakers, and others to forecast the temperatures into the future with confidence allowing for us to better understand how and when global temperatures may rise so that we can be better prepared for the impact of climate change.

Methodology : The Time Series Models that were tested in this project include : 1.Naive , 2.Simple Moving Average (3) , 3.Simple Exponential Smoothing , 4.Holt Winters , 5.ARIMA / SARIMA . For the methodology , each model will fit on the same training data , and then we will use the model fit on the training data and evaluate on the test set and then calculate metrics on the test set to see its true unbiased performance . I will use MAE (Mean Absolute Error) and RMSE (Root mean squared error) as the metrics to choose which model performs best on the test set . MAE is easy to understand and lets us know on average how off is our prediction from the true value and is robust to outliers . RMSE penalizes large errors so especially as our holdout set has a big spike this will let us know if the model struggles with extreme values and swings. In addition , we will conduct exploratory data analysis to get a better understanding of the data before we start modeling to detect if our time series needs differencing , has trend , has seasonality , and to see the auto correlation . And we also conduct residual analysis for all of the models to ensure that the model captures all of the structure in the data and if the model is actually valid for the data in addition to having the best MAE / RMSE on the test set.

Data , exploration insights : The original data is from this kaggle website : https://www.kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data/data?select=GlobalTemperatures.csv . The original data contains average monthly temperature data for 243 countries from 1743 to 2013 . The authors of the dataset have repackaged the data from a newer compilation put together by the Berkeley Earth, which is affiliated with Lawrence Berkeley National Laboratory. The Berkeley Earth Surface Temperature Study combines 1.6 billion temperature reports from 16 pre-existing archives. We can be confident that the recorded data is accurate , especially if we filter for more recent years as the technology to record temperature more accurately has improved with time. I am interested in predicting the global average temperature , so I grouped by country and took the average to get monthly global average temperatures. In order to be more confident that the recorded data is the true temperature , filtered original dataset for 1900 and on . Also there was a new upward trend in temperatures identified when viewing the data at the yearly level starting at year 1956 , so we will filter the time series 1956 and onward and still continue using the data at the monthly level. From viewing the acf plot we can see significant spikes at lag 1 , 2 , 6 , 12 , 18, 24. Intuitively these spikes make sense as to predict the upcoming months weather , the previous 1-2 months will be important . In addition, using the same temperature from the same time but last year or two years ago for the same period will be useful. We can also see the half year and 18 month lags are significant but negatively autocorrelated which also makes sense. From the STL Decomposition of our time series we can see that there is a strong seasonal component as expected as we have monthly global average temperatures . There is an upward trend also as expected due to climate change , more use of fossil fuels with time , and other reasons. Using the kpss test the p value is 0.1 . So we fail to reject the null hypothesis and conclude that the time series is not stationary.

Conclusion , Limitations, Recommendations : Now to conclude , from all the models only Holt Winters and ARIMA / SARIMA were contenders for being the final model for performing best on the test data the rest performed much worse . Both models had the same mae of 0.71 but the Holt Winters model had RMSE of 1.51 while ARIMA / SARIMA had RMSE of 1.60 on the test set. Due to MAE being the same but large difference in RMSE on the test set, I choose the Holt Winters model as the best model to forecast global monthly temperatures. Some limitations are that while forecasting the global monthly temperature is useful, it would be more useful for decision makers to have specific country level forecasts , this would require hundreds of models but would provide more accurate and granular results. In addition , an approach that was not tried in this project was an ensemble model. Given more time , I would have tried the ensemble model by adding weights to the prediction from the holt winters and to the arima / sarima model that add to 1 . Then use grid search with k fold cv on the training set to find the optimal weights i should use then evaluate on the test set and see if that performs better in MAE and RMSE then the original baseline Holt Winters Model.